

Mustafa Alhelawe

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SKILLS

Languages: Scala, JavaScript, TypeScript, Python, C#, Java, Kotlin, Swift, C/C++, Bash

Backend: REST APIs, Distributed Systems, Microservices, JSON, CI/CD, Ktor, KMP, Realtime, RLS/auth

Frontend: React, Node.js, Next.js, Tailwind CSS

Mobile: Android (Jetpack Compose), iOS (SwiftUI), Kotlin Multiplatform

Database: PostgreSQL, Supabase, SQL

AI / ML: PyTorch, Scikit-learn, Reinforcement Learning, LLMs

Tools: Git, Cursor, Claude, Unity, Cypress, Gradle, Jenkins, Linux, Splunk, Plaid, Sentry, Docker

EDUCATION

Rutgers, The State University of New Jersey

New Brunswick, New Jersey

Bachelor of Science in Electrical and Computer Engineering

Class of 2022

WORK EXPERIENCE

Software Engineer

Jersey City, New Jersey

Bank of America Merrill Lynch

January 2023 – Present

- Own backend services within a distributed trade confirmation and allocation system processing millions of daily transactions across high-value institutional trades.
- Built and maintained allocation workflows supporting 10+ distinct trade flows, handling full trade lifecycles including matching, booking, confirmation, amendment, and cancellation.
- Built services to ingest and link orders from multiple upstream systems including FIX-based messaging and file-based workflows, enabling real-time trade matching and confirmation.
- Led migration of 300+ client configurations to a new JSON schema in AMPS SOW, improving query performance and reducing ongoing manual support.
- Replaced biweekly release-dependent YAML/Ansible onboarding flows with a self-service platform, enabling operations teams to onboard clients independently without engineering intervention.
- Achieved improved configuration reliability and reduced production risk by designing approval and diff-review workflows for configuration changes.
- Enhanced search and filtering capabilities, used by operations teams to navigate high-volume datasets more efficiently.
- Led coordination of global teams to ensure zero-downtime rollout of next-generation architecture across 15+ production servers.

Cybersecurity Analyst Intern

East Rutherford, New Jersey

Capri Holdings Limited

June 2022 – August 2022

- Performed vendor risk assessments on various corporate partners, analyzed security policies ensuring best practices were in place when confidential information was at risk.
- Utilized SIEM platforms such as LogRhythm to assist in Level 2 & 3 Information security events/alerts investigations and triage (i.e., detected unauthorized communications with banned IPs in locations that were shut down).
- Reviewed inbound suspicious emails for indicators of compromise with the assistance of CrowdStrike's sandbox tool.

PROJECTS

Splits—IOW Tracker with Auto-Detected Shared Bills

February 2026 – Present

- Built a tri-platform expense-sharing app (web, Android, iOS) with a shared Kotlin Multiplatform module powering recurring bill detection, balance tracking, and split computation. (KMP | Next.js | SwiftUI | Jetpack Compose)
- Designed a Plaid-based recurring transaction engine that clusters merchant + amount patterns across months to automatically detect and split shared bills.
- Developed an idempotent ledger and rule-processing pipeline that safely replays sync retries, recurring jobs, and backfills without duplicate entries. (PostgreSQL | Supabase | Ktor)
- Implemented secure auth and data isolation using AES-256 token encryption, Supabase Row-Level Security, and SECURITY DEFINER RPCs.
- Shipped native Android (Jetpack Compose) and iOS (SwiftUI) clients with shared business logic, OTP auth, and realtime notifications via FCM/APNs.

AI Simulation Platform—Reinforcement Learning

January 2025 – Present

- Built a custom simulation environment in Unity ML-Agents to train autonomous agents using PPO.
- Designed reward systems, state observations, and training pipelines for iterative agent improvement.
- Ran experiments, tuned hyperparameters, and analyzed performance metrics to improve learning stability.
- Developed production-style modular C# environment logic integrated with Python training pipelines.